

A Multi-objective Optimization Problem for Sizing and Siting of Power Systems with the Integration of DG Units, EV Charging Stations, and ESSs

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Abstract—In this paper, a multi-objective optimization problem of sizing and siting power systems with the integration of electric vehicles in a distribution system is addressed. Vehicle-to-grid contribution, charging stations, renewable energy sources in the form of photovoltaic and wind turbines, and energy storage systems are considered in the proposed approach. A mixed-integer linear programming model is presented for this problem, which minimizes the total cost associated with infrastructure development and power generation while considering operational efficiency and sustainability considerations. A strategic placement of such infrastructural arrangements and their sizes can improve grid resiliency and optimally host a high penetration of electric vehicles, along with integrating renewable energy sources, is also proposed. The suitability of the proposed model is proven through a case study presented for an IEEE 24-bus distribution power system, drawing improvements in cost effectiveness, energy efficiency, and renewable energy utilization within the power grid.

Index Terms—Electric vehicles, mixed-integer linear programming, renewable energy sources, Sizing and siting, vehicle-to-grid

NOMENCLATURE

α	Minimum required fraction of total generation from renewables
β	Minimum level of critical load demand to be supplied
ΔT	Time step duration
λ	Weighting factor trade-off between objectives
A	Weighting factor for power losses
B	Weighting factor for generation costs
$I_{line,m}$	Current in transmission line m
$I_{max,m}$	Maximum current in transmission line m

$P_{avail,l}$	Available power for charging station l
$P_{charge,l}$	Power allocated to charging station l
$P_{demand,i}$	Power demand of critical load i
$P_{ESS,j}$	Power exchange with energy storage system j
$P_{EV,j}$	Power exchange with electric vehicle j
$P_{gen,i}$	Power generated by unit i
$P_{load,i}$	Power load at bus i
$P_{PV,i}$	Power generated by photovoltaic system i
$P_{supplied,i}$	Power supplied to critical load i
$P_{Wind,i}$	Power generated by wind turbine i
$Q_{loss,l,t}$	Reactive power loss at line l
R_l	Resistance of line l
$SoC_{j,Max}$	Maximum state of charge of electric vehicle j
$SoC_{j,Min}$	Minimum state of charge of electric vehicle j
$u_{h,i,t}$	Binary variable indicating ESS charging status
V_{max}	Maximum voltage
V_{min}	Minimum voltage
V_n	Voltage at node n
$W_{i,t}$	Squared voltage magnitude at bus i

I. INTRODUCTION

Renewable energy sources have been aggressively developed during the last decades because traditional generation units mostly use fossil fuels, which causes immense pollution. The

scarcity of fossil fuels also urged industries to substitute modern technologies with traditional ones [1,2]. Although renewable energy sources have a wide range of advantages for power generation and distribution, some of their disadvantages are yet to be resolved, especially in power system studies. The high variability of such resources and the fact that these sources are non-dispatchable due to environmental factors lead researchers to take into account these constraints in the operation of power systems [3,4]. The demand side, as well as the generation side, is also an important issue for systems operators since the new types of electric loads have significantly developed that require a higher amount of electricity. Electric vehicles (EVs) and zero emission vehicles (ZEVs), which are a future trend in power system studies due to their positive environmental impacts and improvements in air quality, require a high level of power as a load and, at the same time, have a high capacity for energy storage in vehicle-to-grid (V2G) mode [5]. The sizing and siting of renewable-based distributed generation (DG) units have been widely dealt with in the previous studies. In [6], a comprehensive optimization model for sizing and siting of various types of renewable-based DG units, EV charging stations (EVCSs), and energy storage systems (ESSs) in the distribution system is proposed. Equity is an important factor in locating EVCSs which facilitates the access to the stations by locating them equally for all segments of population [7]. The paper presents two criteria for equity which are policy-based so that the distance between the consumer and the station is less than or equal to a certain value. Paper [8] explores the interplay between transportation and power systems, focusing on decentralized renewable generators and EVs. The research done in [9] introduces an integrated multiple criteria decision-making approach for electric vehicle charging station site selection, crucial for addressing energy and environmental challenges. In [10], an enhanced solution for coordinating multiple facility planning across electric power lines, transportation roads, and charging stations (CSs) is presented. Paper [11] presents an approach for distribution network expansion planning under the $N - 1$ criterion, providing both optimal investment and operation plans. To manage computational complexity, it initially relaxes constraints and then iteratively adds binding contingency constraints using a designed method. This iterative process effectively addresses the distribution network planning problem. The issues that natural disasters like wildfires present to the electrical system are the subject of [12], which focuses on identifying communities at risk. It integrates spatial fire probability data with graph theory assessment techniques, such as betweenness centrality and minimal cut, to evaluate populations at risk of losing power as a result of wildfires. The findings highlight communities at risk of grid separation and essential lines at risk within high-probability burn zones, proving the techniques' efficacy in evaluating grid vulnerability to wildfires. In order to overcome the difficulties associated with large-scale EV integration in power-traffic networks, [13] suggests a coordinated solution for the stochastic security-constrained unit commitment-traffic assignment problem (SCUC-TAP). It uses TAP for traffic

networks and stochastic SCUC for power grid management to examine how EV departure times affect congestion. In order to estimate the demand for electric vehicle charging during evacuations, which are high-demand scenarios due to natural disasters, [14] presents a unique model that is integrated into an algorithmic approach for grid impact prediction. California is extremely at risk of extreme weather conditions and natural disasters including earthquakes, floods, and wildfires, all of which have an enormous adverse effect on property and human life. Adopting ZEVs is a crucial strategy to address these environmental issues. This strategy can particularly assist local governments and communities in improving their emergency preparedness for ZEV-based evacuations [15]. Operator awareness is improved by the geospatial visualization of evacuation charging demand, which highlights crucial areas for proactive grid management. Distribution system planning usually includes a variety of expenses, but sizing and siting pose distinct issues.

In this paper, the optimization of power system infrastructure, focusing on optimal sizing and siting of DGs, photovoltaic (PV) systems, wind turbines (WTs), ESSs, and EVCSs across a 24-bus system, is formulated as a mixed-integer linear programming (MILP) problem. A multi-objective optimization (MOO) approach to minimize power losses while maximizing available generation, ensuring infrastructure diversity, and achieving an optimal energy mix tailored to local conditions and policy goals is employed.

II. FORMULATION AND OPTIMIZATION PROBLEM

This section outlines the mathematical formulation of the MOO for power system planning, aimed at minimizing operational costs, investment in new technologies, and environmental impacts, subject to a variety of system constraints that ensure the reliability and feasibility of the power system operation.

A. Objective function

The objective of the optimization problem is to minimize the power losses, as well as maximize the penetration of energy generation and storage. This multi-objective function reflects the complex trade-offs in power system planning and operation:

The objective function can be formulated as follows:

$$\min Z = w_1 \cdot C_{\text{Losses}} + w_2 \cdot C_{\text{Generation}} \quad (1)$$

where:

- w_1, w_2 are the weighting factors for each objective.
- C_{Losses} represents the cost associated with power losses.
- $C_{\text{Generation}}$ represents the investment and operational costs for DG units, EVCSs, and ESSs.

The objective function is mathematically expressed as follows:

$$\begin{aligned} \min Z = & \sum_t \sum_l A \cdot (\hat{P}_{\text{loss},l,t}) \\ & - B \cdot (P_{\text{DGcap},i} + P_{\text{EVScap},i} + E_{\text{ESScap},i}) \\ & + \lambda \left(\sum_{r \in R} P_{\text{gen},r} / \sum_{i \in I} P_{\text{gen},i} \right) \end{aligned} \quad (2)$$

B. Constraints

The optimization problem is solved by considering different physical and operational constraints of the power system, ensuring that the solutions are both feasible and reliable.

1) *Power Balance*: Guarantees that at every moment and for each bus in the system, the total generated power equals the total demand plus the net exchange with ESSs and EVs, the active and reactive power balance equations for each bus i are given by [6]:

$$P_{G,i,t} - P_{L,i,t} = \sum_{l \in Bi,j} (M_{F,i,l} \cdot P_{r,l,t} + M_{L,i,l} \cdot P_{\text{loss},l,t}), \forall i, t \quad (3)$$

$$Q_{G,i,t} - Q_{L,i,t} = \sum_{l \in Bi,j} (M_{F,i,l} \cdot Q_{r,l,t} + M_{L,i,l} \cdot Q_{\text{loss},l,t} - B_l \cdot M_{W,l,i} \cdot W_{i,t}), \forall i, t \quad (4)$$

2) *DC Power Flow and Voltage Limit*: A simplified DC power flow equation and voltage limits for each bus between minimum and maximum bounds are as follows:

$$P_{ij} \approx \frac{\theta_i - \theta_j}{X_{ij}}, \forall (i, j) \in E \quad (5)$$

$$V_{\min} \leq V_i \leq V_{\max}, \forall i \in N \quad (6)$$

3) *Charging Stations*: Limit the power allocated to CSs, ensuring that the infrastructure supports EV charging without overloading the local grid. The constraint for EVCSs is as follows:

$$0 \leq P_{\text{EV},l,t} \leq P_{\text{avail},l}, \forall l, t \quad (7)$$

$$-\Delta P_{\max} \leq P_{\text{EV},l,t} - P_{\text{EV},l,t-1} \leq \Delta P_{\max}, \forall l, t > 1 \quad (8)$$

$$t_{\text{charge},l} \leq T_{\max}, \forall l \quad (9)$$

Where (8) and (9) are the rate of change of charging power, preventing sudden spikes in demand and maximum charging time, respectively.

4) *ESSs Operational Constraints*: For mitigating the variability of renewable energy and enhancing grid flexibility, these constraints ensure that the charging and discharging power of the ESSs remain within specified limits, the state of energy (SOE) is updated correctly at each time step, and the SOE stays within defined minimum and maximum bounds. The constraints for ESS charging and discharging are as follows:

$$0 \leq P_{\text{ESS,ch},i,t} \leq R_{\text{ESS,ch}}, \quad 0 \leq P_{\text{ESS,dis},i,t} \leq R_{\text{ESS,dis}}, \forall t \quad (10)$$

$$\text{SOE}_{\text{ESS},i,t} = \text{SOE}_{\text{ESS},i,t-1} + C_{\text{ESS}} \cdot P_{\text{ESS,ch},i,t} \cdot \Delta T - P_{\text{ESS,dis},i,t} \cdot \Delta T, \forall t \geq 1 \quad (11)$$

$$\text{SOE}_{\text{ESS,min}} \leq \text{SOE}_{\text{ESS},i,t} \leq \text{SOE}_{\text{ESS,max}}, \forall t \quad (12)$$

5) *State of Charge (SOC)*: Manage the interaction between the grid and electric vehicles, both in terms of charging requirements and the potential for V2G services. The SOC constraints for EV batteries are:

$$\text{SoC}_{j,\text{Min}} \leq \text{SoC}_j \leq \text{SoC}_{j,\text{Max}}, \quad 0 \leq P_{\text{EV},j} \leq \text{Max}P_{\text{EV},j} \quad \forall j \quad (13)$$

Aiming to ensure a certain percentage of the total energy mix is generated from renewable sources, such as wind and PV, and aligning with policy goals or regulatory mandates for sustainable energy, constraints (14), (15), and (16) are proposed.

$$\sum_{i \in \text{Renewables}} P_{\text{gen},i} \geq \alpha \sum_{i \in \text{All Sources}} P_{\text{gen},i} \quad (14)$$

where α represents the minimum required fraction of total generation that must come from renewable sources.

6) *Wind Power Generation*: The constraint for wind power generation can be defined as follows:

$$P_{\text{Wind},i,t} = f(v_{i,t}) \quad (15)$$

Represents the power generated by wind turbines, $P_{\text{Wind},i,t}$, modeled as a function of wind speed, $f(v_{i,t})$.

7) *PV Power Generation*: The constraint for PV power generation can be expressed as follows:

$$P_{\text{PV},i,t} = n_{\text{PV},i} \cdot \eta_{\text{PV}} \cdot \frac{\bar{I}_{\text{ACT},i,t}}{\bar{I}_{\text{REF}}} \cdot P_{\text{rated}} \quad (16)$$

Represents the power generated by PV systems, $P_{\text{PV},i,t}$, calculated based on the number of PV units, their efficiency, actual irradiance, and reference irradiance.

8) *Grid Capacity and Safety Constraints*: Ensure the physical and safety limits of the power system are not exceeded, including transmission line capacities and voltage regulations:

$$I_{\text{line},m} \leq I_{\text{max},m}, \quad V_n \in [V_{\min}, V_{\max}] \quad \forall m, n \quad (17)$$

9) *Investment and Siting*: These constraints use binary and continuous variables to model decisions about deploying new assets. These constraints balance economic, technical, and environmental considerations to ensure solutions that are cost-effective, sustainable, reliable, and resilient.

1. Binary Variable: x_i (1 if an asset is deployed at location i , 0 otherwise).

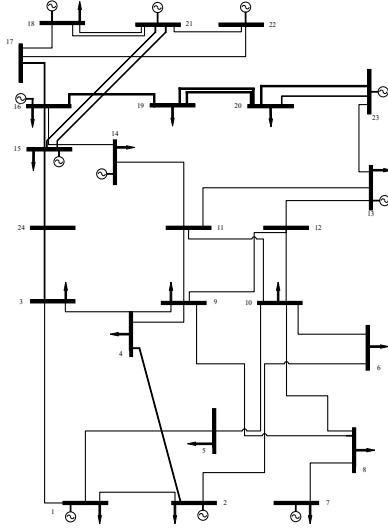


Fig. 1. IEEE RTS 24-Bus System.

2. Continuous Variable: y_i (size or capacity of the asset at location i).

These constraints ensure that the proposed optimization model is comprehensive and takes into account various aspects of the power system's operation, investment, and resilience, particularly in the context of ZEV-based evacuation during natural disasters.

III. SIMULATION RESULTS

Incorporating the system and specifications provided in [16] and employing the IEEE RTS 24-bus system as the test case, as shown in Fig. 1, the optimization problem of power system infrastructures is investigated. In this regard, a multi-objective optimization framework has been applied to this well-established test system, introducing varied and realistic capacity ranges alongside investment and operational costs for each infrastructure type. This approach not only reflects a real-world application of power system enhancement but also allows for a detailed analysis of infrastructure deployment across the network. The candidate infrastructures, outlined in Table I and II showcasing the specifications of these infrastructures and the detailed allocation of new capacities across the test system, reveal a strategic distribution of energy sources that emphasize economic efficiency and grid reliability. Notably, the optimization yields a diversified energy portfolio where renewable energy sources, ESSs, and CSs complement each other, ensuring a robust response to demand fluctuations and infrastructure constraints. The objective function of the optimization problem has been defined in Pyomo, an open-source library in Python, and then the mentioned optimization problem, subject to several constraints, has been solved in CPLEX, a commercial solver.

Through this case study, the proposed planning framework's applicability and effectiveness are demonstrated, underscoring

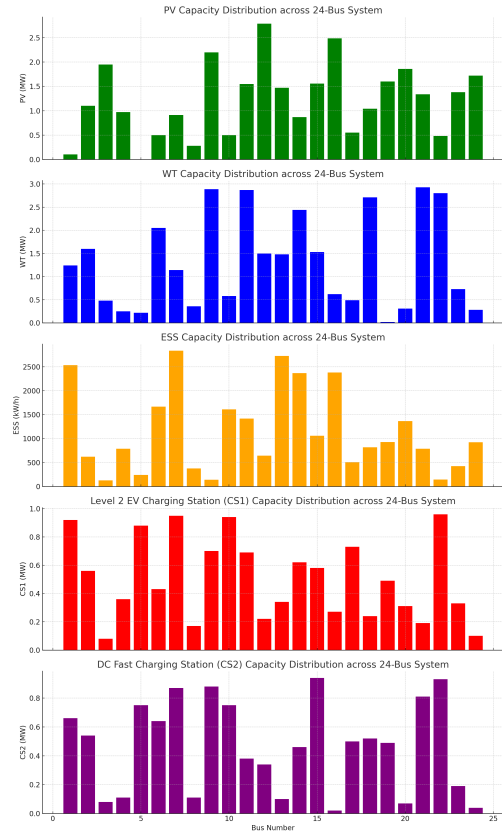


Fig. 2. Distribution of PV, WT, ESS, and CSs across the 24-bus system.

its potential to guide strategic infrastructure development. This approach provides valuable insights into optimizing power system expansions by balancing investment and operational expenditures. The final expansion plan is determined by aligning with decision-makers' preferences for various objectives. The 24-bus system's infrastructure allocation is shown in Table III. Figure 2 shows the tailored infrastructure allocation for each bus in the 24-bus system, incorporating renewable sources to ensure resource optimization and recovery capacity. The results demonstrate an optimized distribution that minimizes power losses, enhances grid reliability, and balances the placement of ESSs and EV charging stations to support grid flexibility and capacity. Overall, the integration of these resources effectively optimizes grid performance and resilience.

IV. CONCLUSION

This paper successfully shows the effectiveness of an MILP-based optimization framework for strategic sizing and siting of power system infrastructure with the integration of EVs and renewable energy sources. Through a case study of a 24-bus distribution power system, it points to possible ways of significantly improving grid efficiency, cost reduction, and sustainability. The proposed model optimizes not only the cost and power generation aspect but also makes it feasible to integrate more EVs using concepts of V2G, renewable energy sources, and energy storage systems. These findings

TABLE I
DESIRABLE INFRASTRUCTURES AND THEIR SPECIFICATIONS

Infrastructure	Investment Cost (k\$/MW)	Operation Cost (k\$/MW/year)	Capacity (MW)	Efficiency (%)	Lifetime (years)
PV System	1000	10	2	18	25
WT	1500	20	2	35	20
ESS	500	15	5	90	15
Level 2 EV Charging Station (CS ₁)	500	5	0.022	95	10
DC Fast Charging Station (CS ₂)	1000	10	0.35	95	10

TABLE II
DEGRADATION RATE AND CO₂ EMISSIONS REDUCTION OF INFRASTRUCTURES

Infrastructure	Degradation Rate (%/year)	CO ₂ Emissions Reduction (tons/MW/year)
PV System	0.5	80
WT	0.3	60
ESS	2	0
Level 2 EV Charging Station (CS ₁)	1	5
DC Fast Charging Station (CS ₂)	1	10

TABLE III
SIZING AND SITING RESULTS OF NEW INFRASTRUCTURES FOR 24-BUS SYSTEM

Bus	PV (MW)	WT(MW)	ESS (kW/h)	CS ₁ (MW)	CS ₂ (MW)
1	0.10	1.24	2531.76	0.92	0.66
2	1.10	1.60	622.97	0.56	0.54
3	1.95	0.48	128.27	0.08	0.08
4	0.97	0.25	785.30	0.36	0.11
5	0.00	0.22	241.56	0.88	0.75
6	0.50	2.05	1668.18	0.43	0.64
7	0.91	1.14	2833.65	0.95	0.87
8	0.28	0.36	375.85	0.17	0.11
9	2.20	2.89	140.75	0.70	0.88
10	0.50	0.58	1608.22	0.94	0.75
11	1.55	2.87	1415.89	0.69	0.38
12	2.79	1.50	643.25	0.22	0.34
13	1.47	1.48	2726.66	0.34	0.10
14	0.87	2.44	2368.66	0.62	0.46
15	1.56	1.53	1058.95	0.58	0.94
16	2.49	0.62	2380.31	0.27	0.02
17	0.55	0.49	506.15	0.73	0.50
18	1.04	2.71	818.44	0.24	0.52
19	1.60	0.02	928.91	0.49	0.49
20	1.86	0.31	1366.91	0.31	0.07
21	1.34	2.93	788.55	0.19	0.81
22	0.48	2.80	145.64	0.96	0.93
23	1.38	0.73	424.85	0.33	0.19
24	1.72	0.28	922.77	0.10	0.04

underline the increasing importance of using advanced optimization techniques to plan and upgrade power systems for modern energy network demands. Future works may include scalability of the model to larger power systems, transportation networks modeling and traffic in case of natural hazards to be evacuated, and model resilience in different grid conditions.

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